



Instructor: Date:

Learner: Regional Branch:

Program: Center:

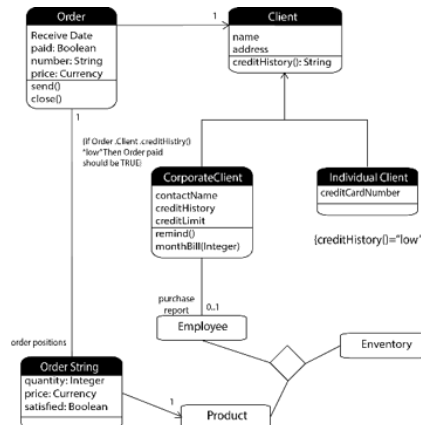
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Let's Start!

What's UML? Let's find out!



Learning Outcome:

Describe common activities at work using short sentences in familiar spoken and written situations.
Interpret the requirements report to determine the technological needs in managing information, based on rules and protocols established by the company.

Ask your partner:

- How do you think programmers build software systems?
- How do programmers organize information?

Can Do Statements:

- I can give detailed information about UML.
- I can explain briefly how to use UML to co-workers.
- I can recognize specific aspects related to UML.



Let's Explore!

This is a student's presentation about UML.



Instructions: Read the text below.

Instructor: Pedro, can you explain UML, please?

Pedro: OK. As you know, building software systems is not an easy task. So UML helps programmers to define and build these systems. Besides, using UML allows creators to define, visualize, build, and document different parts related to software. Creating an application, for example, can be easier if you use UML because it uses Meta object facility which is very precise. Additionally, deploying multiple platforms is possible when programmers use an entire executable UML and different technologies.

Instructor: Good answer! What about the diagrams? Can you mention them?

Pedro: You can find two categories. These are structure and behavior diagrams. Through the first one, you can represent structural information. For example, in the component diagram, you can see the division of components in a software system. In the class diagram, you can see relationships and attributes among the classes. You can also recognize some other specific types of diagrams in the manual; for example, deployment, composite structure, object, and package diagrams. On the other hand, through behavior diagrams, you cannot exemplify structural information because it only displays the functionality of the software. The most common types of these diagrams are Activity, Use Case, UML State Machine, Communication, Timing, Interaction, Overview, and Sequence Diagrams.

Instructor: OK, great job! Can you tell your co-workers how to use UML?

Pedro: Well, the first thing you need to know is about the views of a system model. So, using objects, attributes, operations, and relations is necessary to emphasize the static structure of systems. In this way, you can see that the static view includes class diagrams and composite structure diagrams. In contrast, using modeling tools can help the users to simplify and represent the dynamic view. This enables collaboration among objects and modifies internal states of objects through activity, sequence, and state machine diagrams.



Let's Practice!

Pedro explains some information about UML.

Instructions: Read the following sentences based on the text. In front of each sentence write True or False.

Building software systems is not an easy task. So UML helps programmers to define and build these systems. Besides, using UML allows creators to define, visualize, build, and document different parts related to software. Creating an application, for example, can be easier if you use UML because it uses Meta object facility which is very precise. Additionally, deploying multiple platforms is possible when programmers use an entire executable UML and different technologies.

1. Building software systems is an easy task. _____
2. Creating an application can be really simple if you use UML because it employs Meta object facility. _____
3. Deploying multiple platforms is not probable when programmers employ a complete executable UML. _____

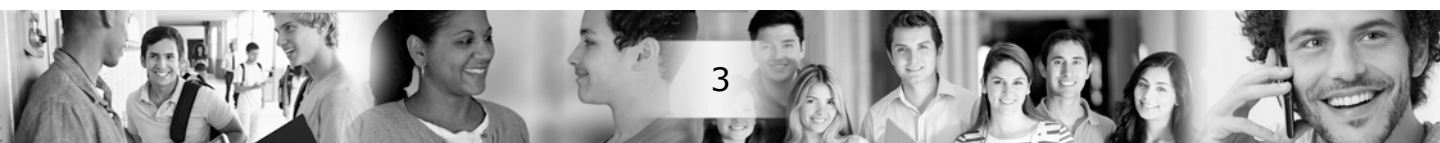


Let's Practice!

My co-workers need some explanation about this topic.

Instructions: Read the sentences. Write the missing words in the blanks. Use: cannot represent, can find, can recognize.

1. You _____ two categories in UML these are structure and behavior diagrams. Through the first one, you can represent structural information.
2. We _____ some other specific types of diagrams in the manual; they are deployment, composite structure, object, and package diagrams.
3. Through behavior diagrams you _____ structural information because it shows specifically the functionality of software. These are Activity, Use Case, and UML State Machine.





Extension Activity

Let's explain to your classmates how to use UML!

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for more information about UML.
- Prepare a presentation on a poster to explain to your classmates how to use UML.
- Give a detailed description about it. Use graphics, pictures, and examples to represent them.
- Use the grammar and vocabulary you studied in class.
- Share your work with your classmates and teacher.



Target Words

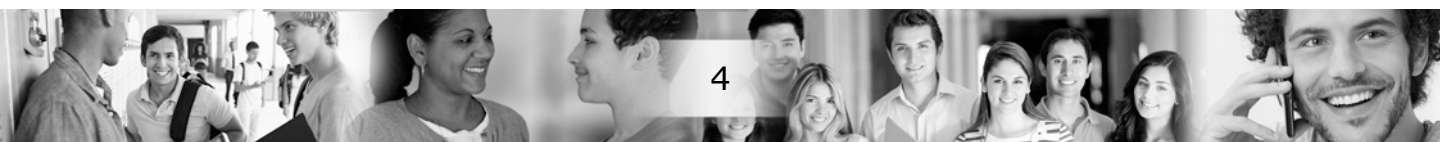
- **executable** (adjective): when something is ready for execution
e.g. Additionally, deploying multiple platforms is possible when you use an entire executable UML and different technologies.
- **relationship** (noun): a connection between different elements
e.g. In the class diagram, you can see attributes and relationships among the classes.
- **functionality** (noun): it refers to the tasks a computer or any other electronic system does
e.g. Through behavior diagrams you cannot exemplify structural information because it only displays the functionality of software.

Self-Assessment

Yes
No

- I give detailed information about UML. ☐ ☐
- I prepare a presentation to explain briefly how to use UML. ☐ ☐
- I recognize specific aspects related to UML and I describe them through graphics, pictures, and examples. ☐ ☐
- I use the grammar and vocabulary I learned. ☐ ☐
- I share my poster with my class. ☐ ☐

- **enable** (verb): to allow or permit
e.g. This enables collaboration among objects and modifies internal states of objects through activity, sequence, and state machine diagrams.
- **co-worker** (noun): a person who works in the same field as you
e.g. OK, great job! Can you tell your co-workers how to use UML?
- **dynamic** (adjective): something that has movement and is active
e.g. In contrast, using modeling tools can help the users to simplify and represent the dynamic view.





Learning Points

Gerunds as Subjects.

You can use **gerunds as the subject** of a sentence.

In English, you can find that some verbs end with the **suffix -ing**; these are called gerunds. In this point, it is important to know that not all the verbs that have -ing take the form of the present continuous. You can also use the -ing form of a verb as the subject of a sentence.

This is one of the structures you can use when including gerunds in a sentence:

Gerund (subject) + verb in singular form + complement

Examples:

- **Deploying multiple platforms** is possible when programmers use an entire executable UML.
- **Developing a software system involves** many different processes.

Regarding the construction of gerunds, take a look to the following chart:

SPELLING RULES FOR GERUNDS		
Type of Verb	Rule	Examples
1. Verb + -ing	You just add -ing	build – building represent – representing design – designing
2. Verb -e + -ing	You drop -e and then add -ing	move – moving create – creating use – using
3. Consonant + vowel + consonant +-ing	You double the last consonant Exceptions: • You don't double the consonants: y, w, and x. • With a two or more syllable verb, you double the last consonant only if the stress goes on the last syllable.	run – running stop – stopping skip – skipping display – displaying draw – drawing fix – fixing begin – beginning admit – admitting refer – referring

Examples

- *Building software systems is not an easy task.*
- *Creating an application, for example, can be easier if you use UML because it uses meta object facility which is very precise.*
- *Using modeling tools can help the users to simplify and represent the dynamic view.*





Learning Points

UML Structure Diagrams.

Structure diagrams are types of diagrams within the Unified Modeling Language (UML) that represent the static structure of a system and the relationship between its parts.

The elements of a structure diagram represent important aspects of a system. These diagrams are not on function of time and they do not show the dynamic behavior of the system. Through a UML structure diagram, it is possible to see the relationship of the system elements' behaviors.

The main types of structure diagram are the following:

Class diagram	Class diagram shows the static structure of a system. It also depicts a set of static elements, called "classes" along with its contents and relationships.
Object diagram	You can say that an object diagram is a kind of snapshot of a class diagram which displays the detailed static structure of the system at a particular time, in addition to test the accuracy of class diagrams.
Component diagram	It shows a set of components and their relationships. These components can be classes, interfaces or collaborations. It allows to describe systems with service-oriented architecture (SOA), representing the view of the implementation of a system. In other words, it is used to display the application.
Deployment diagram	Deployment diagrams are a set of nodes and their relationships. Nodes are physical entities where the components are used. It is used to show the system architecture for the implementation of different deployment targets.